SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05 Total Marks:50

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem *(2)*	Application of Engineering Concepts *(2.5)*	Solution Design *(2.5)*	Use of Tools *(2)*	Analysis *(1)*	*Total(10)*
Weightage	*20%*	25%	25%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I K. Venkata Karthik / 192372092 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Venkata Karthik

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:
Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

Answer:

1. Understanding of the Industry Problem

A supermarket has customers with different purchasing behaviors. Some customers visit frequently but spend less, while others visit rarely but spend more. Identifying these customer groups helps the supermarket design **targeted marketing strategies**.

The dataset contains:

- Customer Visits/Month
- Average Spend
- Number of Items Bought

2. Application of Clustering

A suitable algorithm is **K-Means Clustering**.
Before clustering, the numerical attributes should be **standardized** because average spending has a much larger numerical scale than visits or items.

The customer data can be represented as:

Customer	Visits	Avg Spend	Items
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

For this small dataset, **K = 2** can be selected to identify two broad customer segments.

3. Solution Design

Cluster 1 – Frequent/Regular Customers

- C1
- C3
- C5

These customers visit frequently, purchase many items, but have relatively moderate spending.

Cluster 2 – High-Value Occasional Customers

- C2
- C4

These customers visit less frequently but have considerably higher average spending.

4. Targeted Marketing

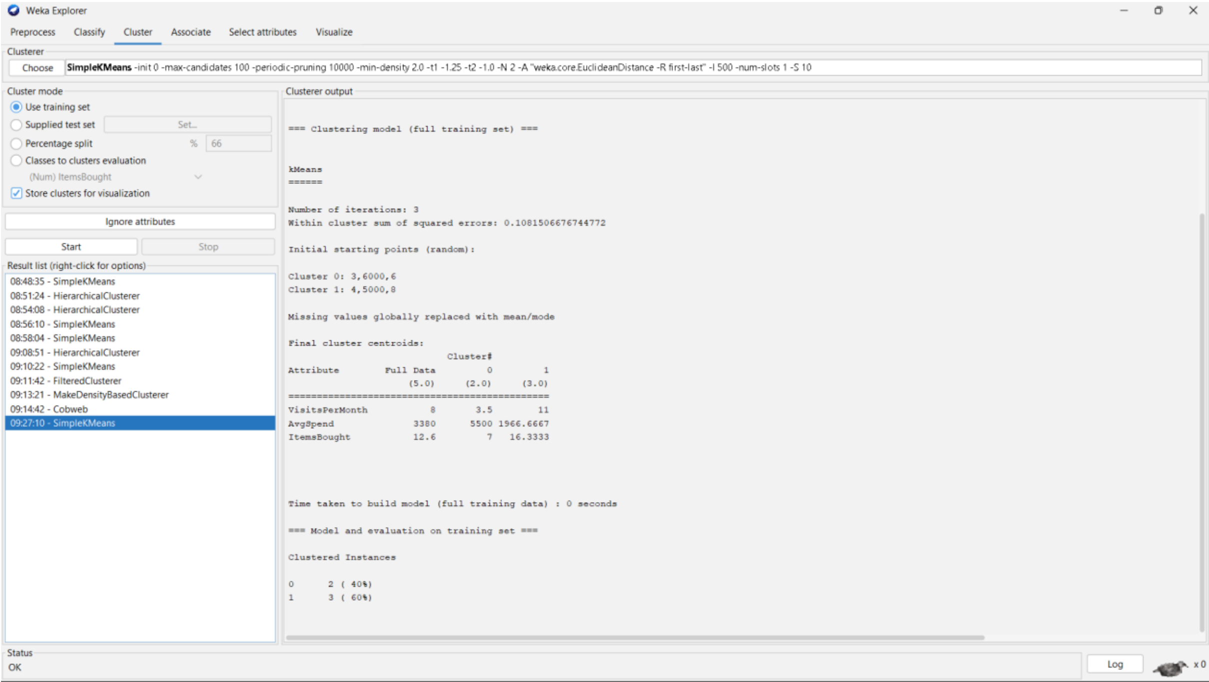
Customer Segment	**Marketing Strategy**
Frequent customers	Loyalty points, weekly discounts, personalized coupons
High-value customers	Premium offers, high-value product recommendations, exclusive deals

5. Analysis

K-Means helps the supermarket understand customer behavior without manually examining every customer. The segmentation can be visualized using a **scatter plot of Visits vs Average Spend**, with different colors representing clusters.

Conclusion:

Clustering allows the supermarket to identify meaningful customer segments and develop personalized marketing campaigns, increasing customer engagement and sales.



Q2. Dataset: Credit Card Transactions

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:
Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

Answer:

1. Understanding of the Industry Problem

Banks and financial institutions need to identify unusual transactions quickly. Fraudulent transactions may differ from normal transactions in terms of **amount, location, and transaction time**.

Dataset:

Transaction	Amount ₹	Location	Time
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

2. Application of Clustering

A clustering-based approach can use **K-Means** or, preferably, **DBSCAN** for anomaly detection.

DBSCAN is useful because it identifies dense groups of normal transactions and treats isolated points as potential anomalies.

For numerical processing:

- Transaction amount → numerical
- Time → numerical
- Location → converted using suitable encoding

3. Solution Design

The transactions show two obvious behavioral patterns.

Normal transaction cluster:

- T1 – ₹200, Chennai, 10 AM
- T3 – ₹150, Chennai, 11 AM
- T5 – ₹180, Chennai, 12 PM

These transactions have similar amounts, location, and time.

Potential unusual transactions:

- T2 – ₹5000, Delhi, 2 AM
- T4 – ₹7000, Mumbai, 1 AM

These transactions are far from the normal cluster.

4. Fraud Detection Process

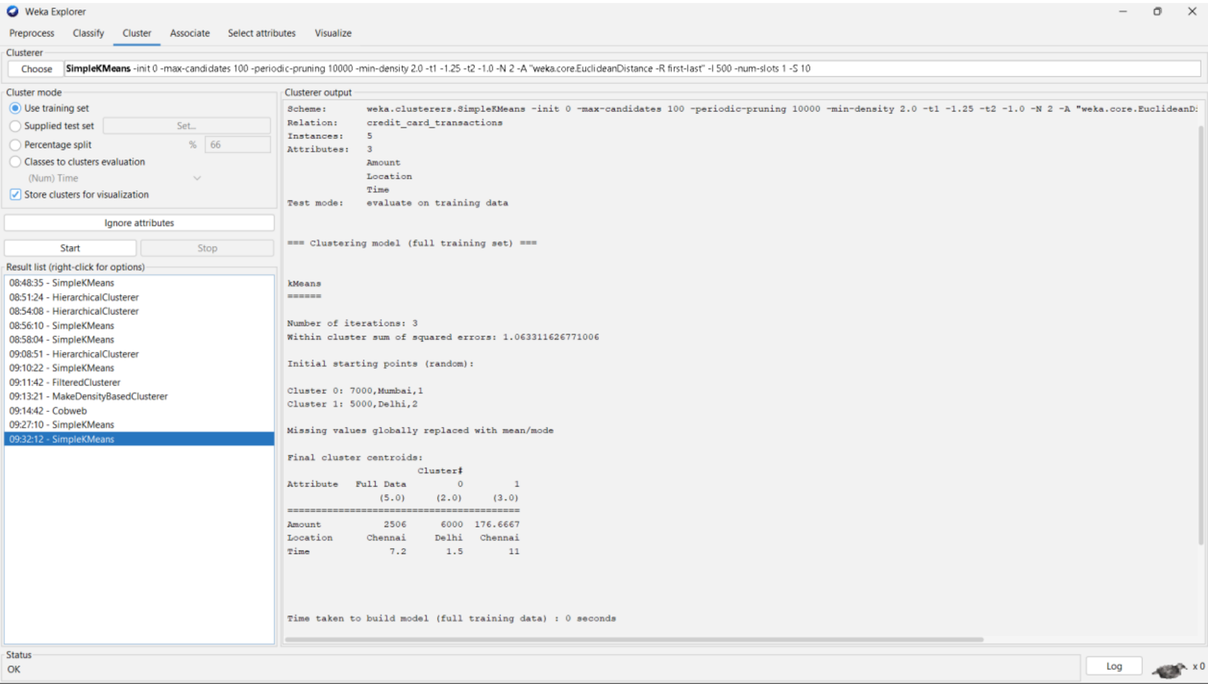
Transaction Data
↓
Data Preprocessing
↓
Feature Scaling & Encoding
↓
Clustering / DBSCAN
↓
Identify Dense Normal Group
↓
Detect Outliers
↓
Fraud Risk Score
↓
Alert Bank / Customer

5. Analysis

The clustering system can flag transactions that are significantly different from normal behavior. However, clustering alone should **not automatically classify a transaction as fraud**. It should generate an alert for further verification.

Conclusion:

Clustering can support real-time fraud detection by identifying unusual transaction patterns and reducing the number of suspicious transactions requiring manual investigation.



Q3. Dataset: Telecom Usage

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

Answer:

1. Understanding of the Industry Problem

Telecom companies want to prevent customers from leaving their services. Customers with low usage and low recharge activity may have a greater risk of becoming inactive.

Dataset:

User	Call Duration	Data Usage GB	Recharge ₹
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

2. Application of Clustering

K-Means clustering can group telecom users according to:

- Call duration
- Data usage
- Recharge amount

Since these features have different scales, they should first be standardized.

3. Solution Design

A suitable segmentation is:

High-usage customers

- U1
- U3
- U5

These users have high call duration, high data usage, and higher recharge values.

Low-usage customers

- U2
- U4

These users have lower usage and lower recharge values and can be considered **potential churn-prone customers**.

4. Retention Strategy

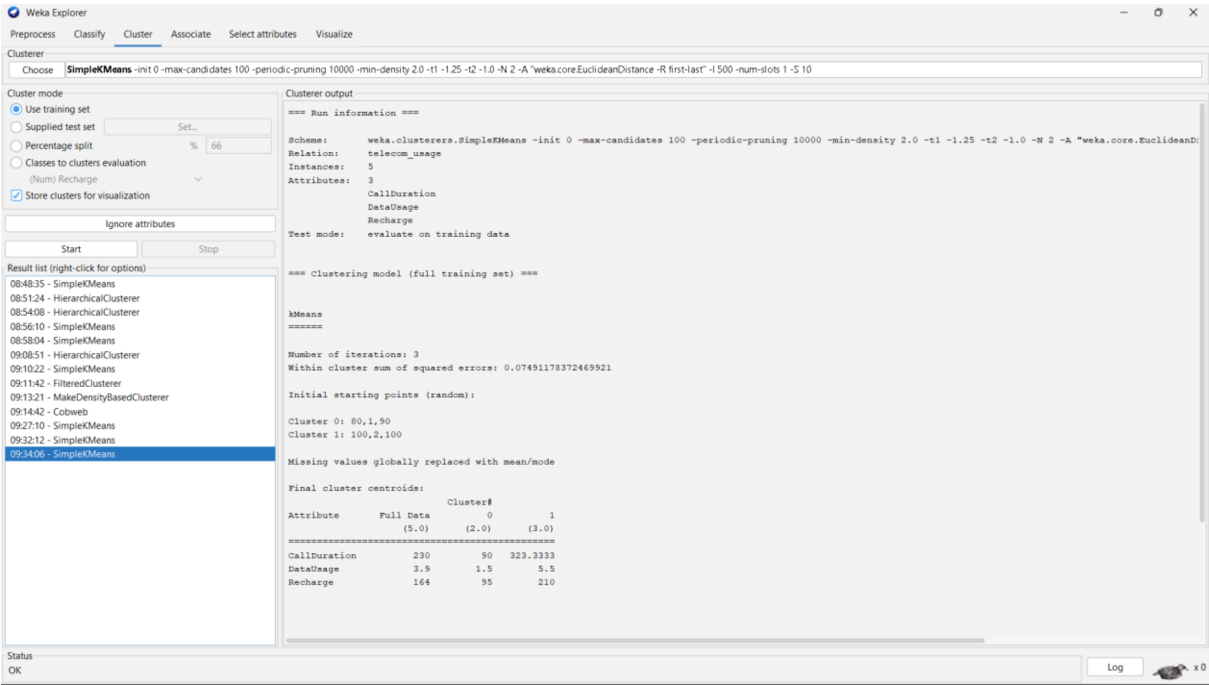
Segment	Strategy
High usage	Premium plans, additional data benefits
Low usage / potential churn	Special recharge offers, free data, personalized plans
Medium usage	Upgrade offers and usage-based packages

5. Analysis

Clustering helps telecom companies identify customer groups based on actual usage behavior. Low-usage clusters can be monitored over time. If usage continues to decrease, the company can send retention offers.

Conclusion:

Clustering provides a data-driven method for identifying potentially churn-prone customers and allows telecom companies to take preventive retention actions.



Q4 . Dataset: E-commerce Behavior

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:
Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Answer:

1. Understanding of the Industry Problem

E-commerce platforms have thousands of users with different browsing and purchasing behaviors. A recommendation system should provide products that are relevant to each user's interests.

Dataset:

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

2. Clustering Technique

K-Means clustering is suitable because the dataset contains numerical behavioral features.

Features:

- Number of product views
- Number of clicks
- Number of purchases

3. Solution Design

The users can be divided into behavioral groups.

Highly engaged users

- U1
- U3
- U5

They have high views, clicks, and purchases.

Low-engagement users

- U2
- U4

They have relatively fewer views, clicks, and purchases.

4. Recommendation System

The cluster information can be combined with product preferences.

For example:

User Activity

↓

Views + Clicks + Purchases

↓

K-Means Clustering

↓

Identify User Segment

↓

Analyze Products Preferred by Cluster

↓

Generate Personalized Recommendations

Highly engaged users can receive:

- Similar product recommendations

- Premium products
- Frequently purchased products
- Cross-selling recommendations

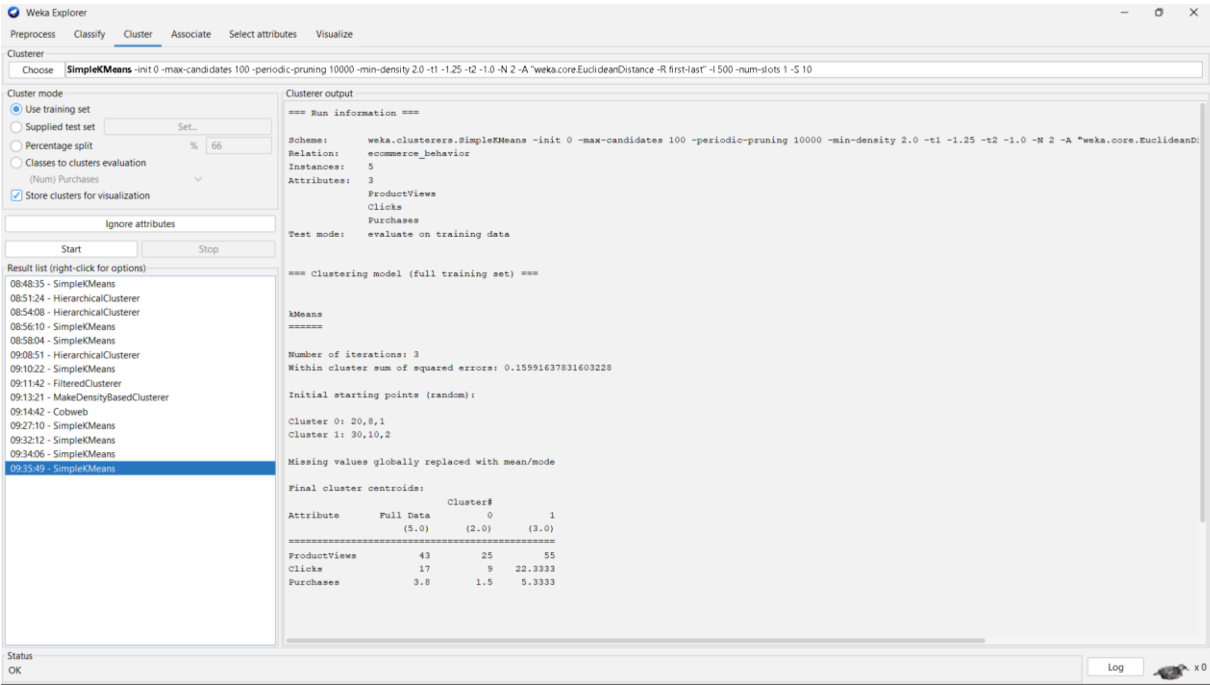
Low-engagement users can receive:

- Popular products
- Discounted products
- Personalized introductory offers

5. Analysis

Clustering helps the recommendation system understand groups of users with similar behavior. This can improve personalization and potentially increase click-through rates and purchases.

Conclusion:
K-Means clustering can divide e-commerce users into meaningful behavioral segments, allowing the recommendation engine to provide more relevant products.



Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25

L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:
Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

Answer:

1. Understanding of the Industry Problem

Traffic congestion causes:

- Increased travel time
- Fuel consumption
- Vehicle emissions
- Delays
- Reduced transportation efficiency

Traffic authorities can use clustering to identify locations with similar traffic characteristics.

Dataset:

Location	Vehicle Count	Avg Speed km/h	Delay min
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

2. Application of Clustering

K-Means clustering can be applied using:

- Vehicle count
- Average speed
- Delay

The features should be standardized before clustering.

3. Solution Design

The locations can be grouped into:

Low-congestion cluster

- L1
- L3
- L5

These locations have:

- Lower vehicle counts
- Higher average speeds
- Lower delays

High-congestion cluster

- L2
- L4

These locations have:

- High vehicle counts
- Low average speeds
- High delays

4. Traffic Management Strategy

Zone	Characteristics	Action
Low congestion	Low traffic, high speed	Normal monitoring
High congestion	High traffic, low speed, high delay	Traffic signal optimization, route diversion, additional monitoring

5. Analysis

Clustering provides a simple way to identify areas with similar traffic conditions. Traffic authorities can visualize clusters on a **city map** using different colors.

For example:

Traffic Data
↓
Preprocessing
↓
Feature Scaling
↓
K-Means Clustering
↓
Low / Medium / High Congestion
↓
Traffic Visualization
↓
Traffic Management Decisions

Conclusion:

Clustering can help transportation authorities identify congestion zones and implement appropriate traffic-management strategies such as signal optimization, route diversion, and real-time monitoring.